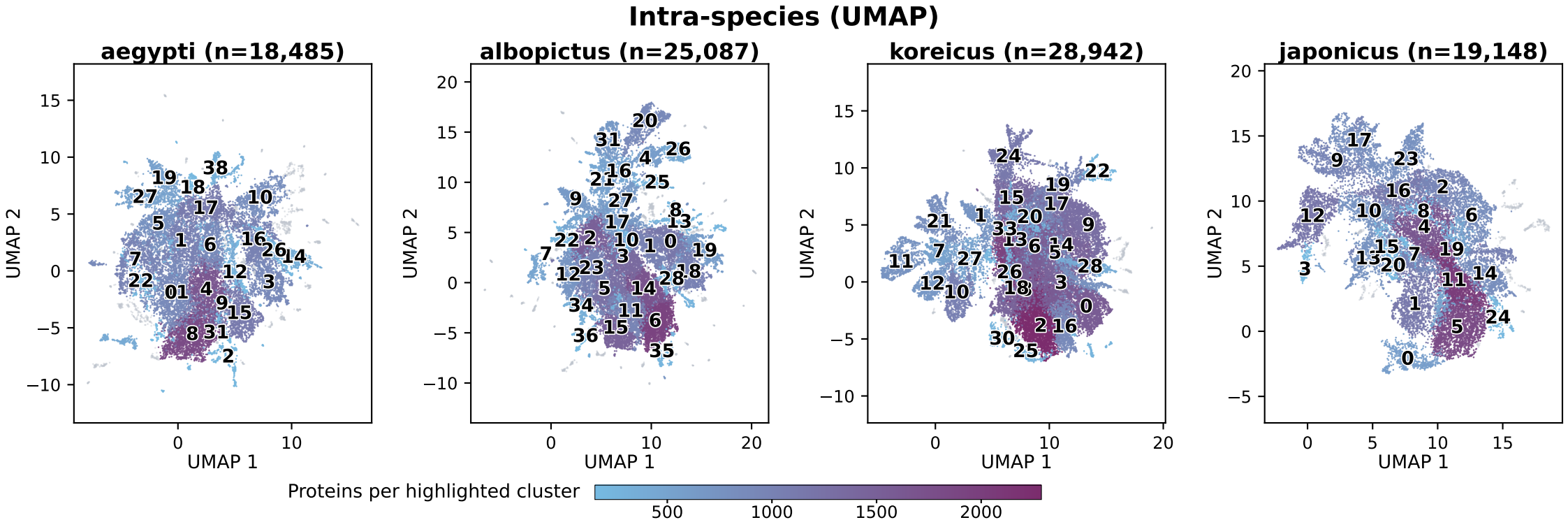
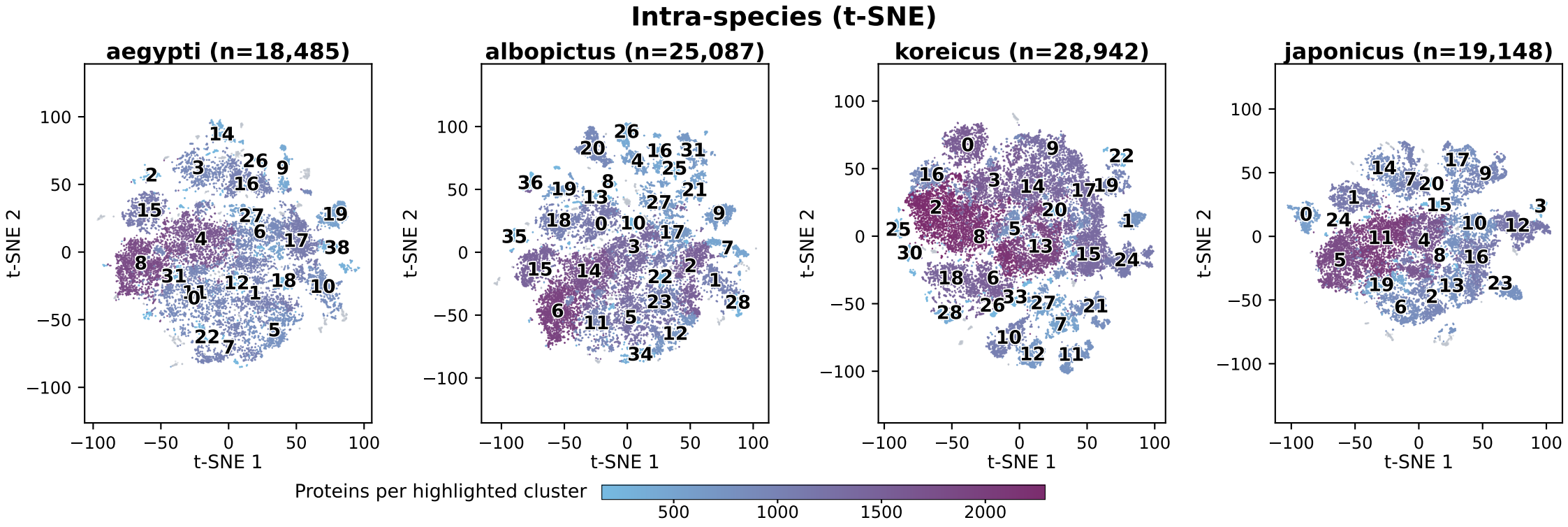
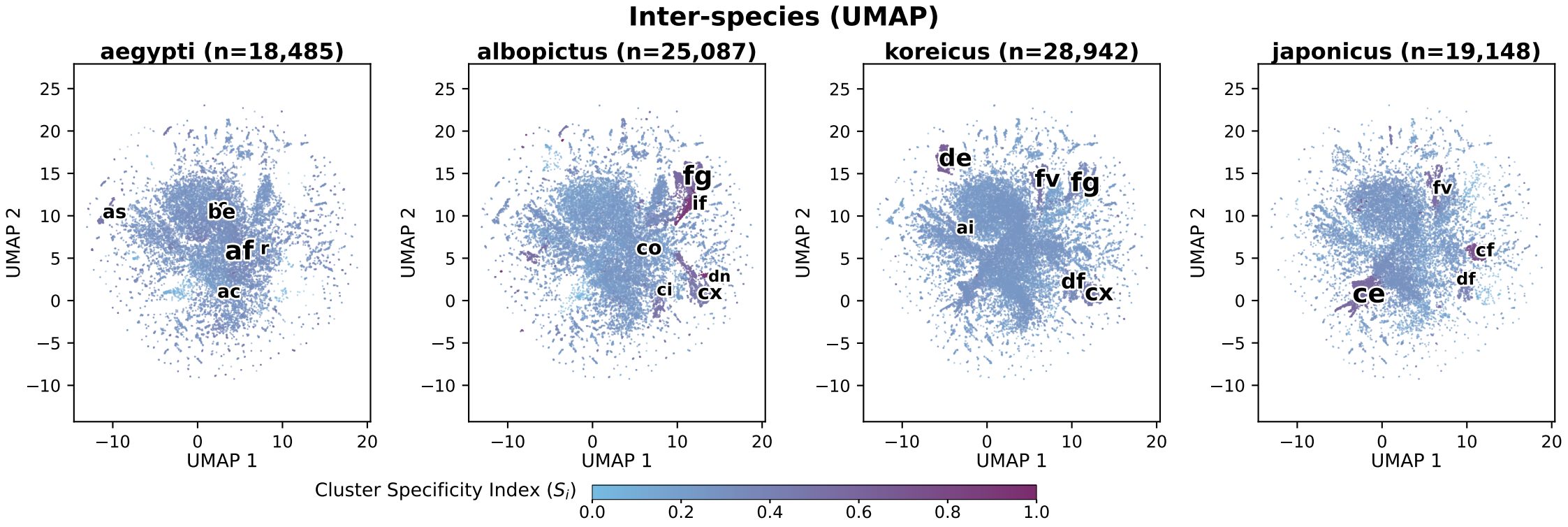
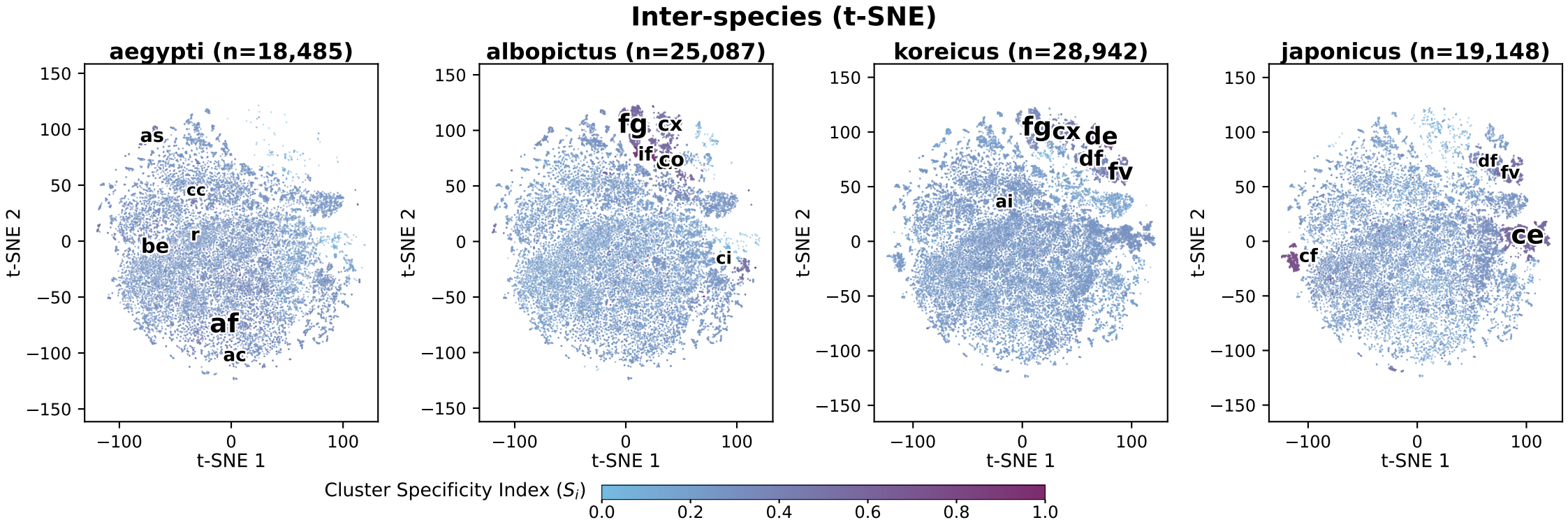
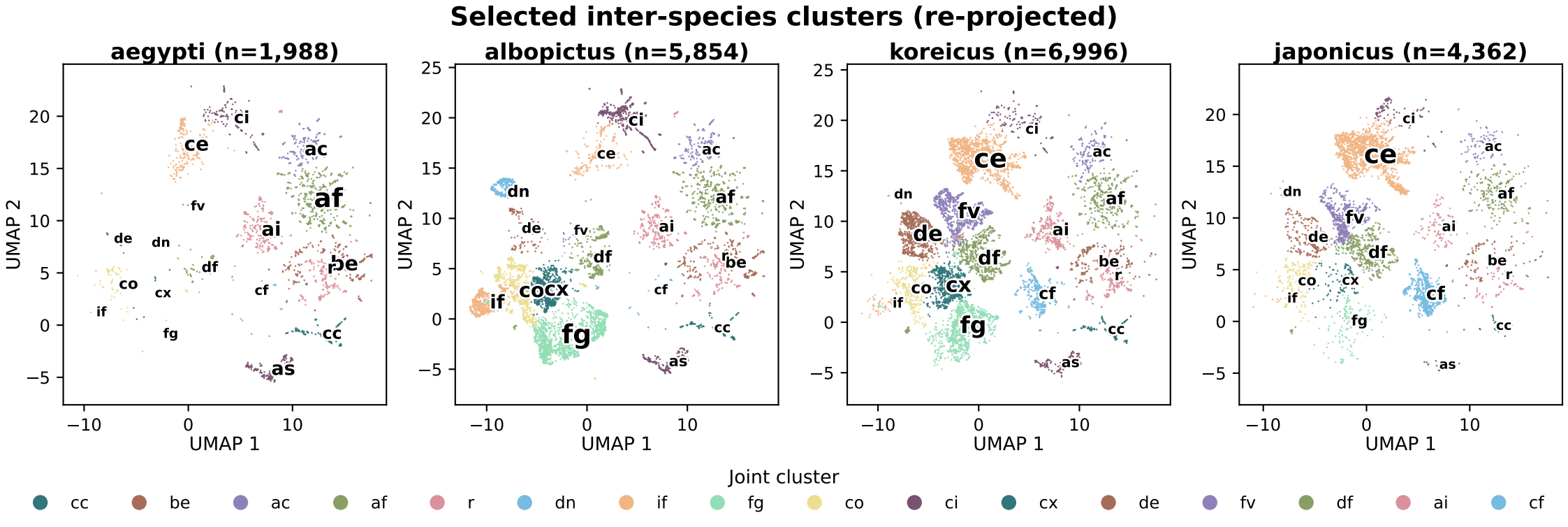

Intra-species (UMAP)
aegypti (n=18,485)
albopictus (n=25,087)
koreicus (n=28,942)
japonicus (n=19,148)
UMAP 1
UMAP 2
Proteins per highlighted cluster
500
1000
1500
2000


Intra-species (t-SNE)
aegypti (n=18,485)
albopictus (n=25,087)
koreicus (n=28,942)
japonicus (n=19,148)
t-SNE 1
t-SNE 2
Proteins per highlighted cluster
500
1000
1500
2000


Inter-species (UMAP)
aegypti (n=18,485)
albopictus (n=25,087)
koreicus (n=28,942)
japonicus (n=19,148)
UMAP 1
UMAP 2
as
be
af
r
ac
fg
if
co
ci
dn
cx
de
fv
fg
ai
df
cx
fv
cf
df
ce
Cluster Specificity Index (S_i)
0.0
0.2
0.4
0.6
0.8
1.0


Inter-species (t-SNE)
aegypti (n=18,485)
albopictus (n=25,087)
koreicus (n=28,942)
japonicus (n=19,148)
t-SNE 1
t-SNE 2
as
cc
be
r
af
ac
fg
cx
if
co
ci
ai
de
df
fv
cf
ce
Cluster Specificity Index (S_i)
0.0
0.2
0.4
0.6
0.8
1.0


Selected inter-species clusters (re-projected)
aegypti (n=1,988)
albopictus (n=5,854)
koreicus (n=6,996)
japonicus (n=4,362)
UMAP 1
UMAP 2
Joint cluster
cc
be
ac
af
r
dn
if
fg
co
ci
cx
de
fv
df
ai
cf

Intra- to Inter-species cluster fluxes

aegypti (n=18,485, k=59)

8
4
15
17
1
16
7
11
3
0
5
9
22
12
31

as (191)
cc (113)
be (233)
ac (177)
af (427)
r (158)

albopictus (n=25,087, k=67)

3
5
0
1
20
17
31
27
9
4
21
7
16
26
22
25

dn (257)
if (557)
fg (1,240)
co (667)
ci (377)
cx (661)

koreicus (n=28,942, k=43)

8
6
14
15
10
21
11
12
7
27

de (752)
fv (717)
cx (700)
fg (897)
df (559)
ai (310)

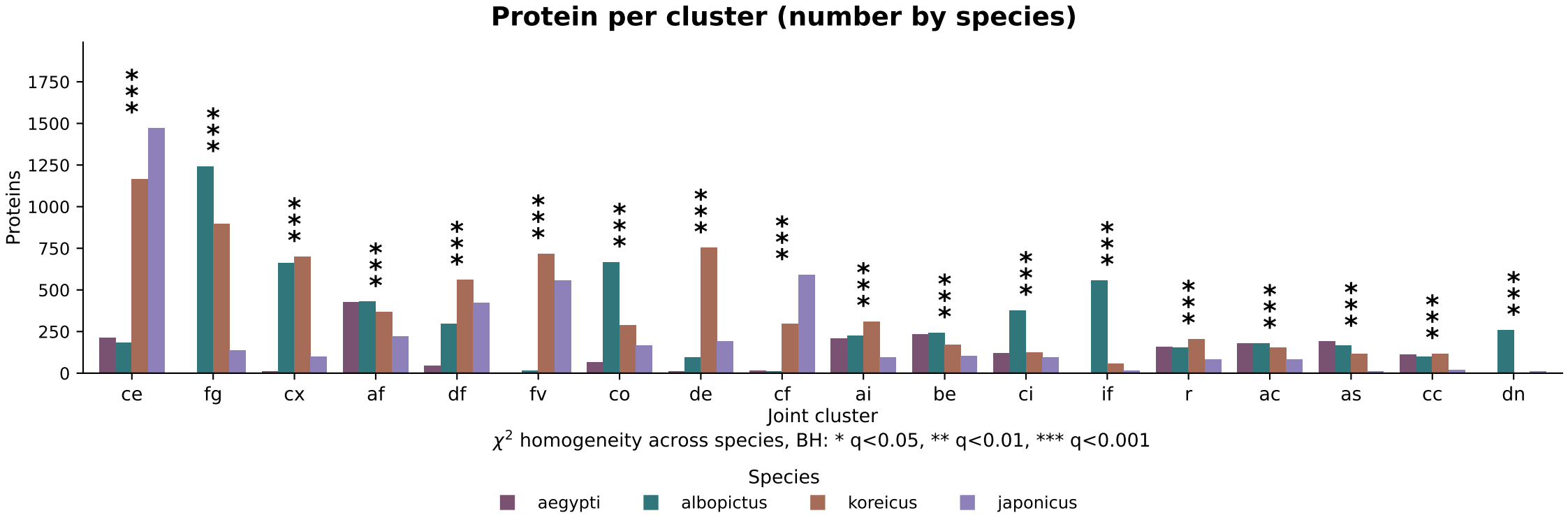
japonicus (n=19,148, k=35)

11
5
4
12
16
9
17
13
10
0
3

cf (591)
ce (1,472)
fv (556)
df (420)

Cluster Specificity Index (S_i); black border: $S_i \geq 0.80$

0.0 0.2 0.4 0.6 0.8 1.0


Protein per cluster (number by species)
Proteins
0
250
500
750
1000
1250
1500
1750
ce
fg
cx
af
df
fv
co
de
cf
ai
be
ci
if
r
ac
as
cc
dn
Joint cluster
χ² homogeneity across species, BH: * q<0.05, ** q<0.01, *** q<0.001
Species
aegypti
albopictus
koreicus
japonicus